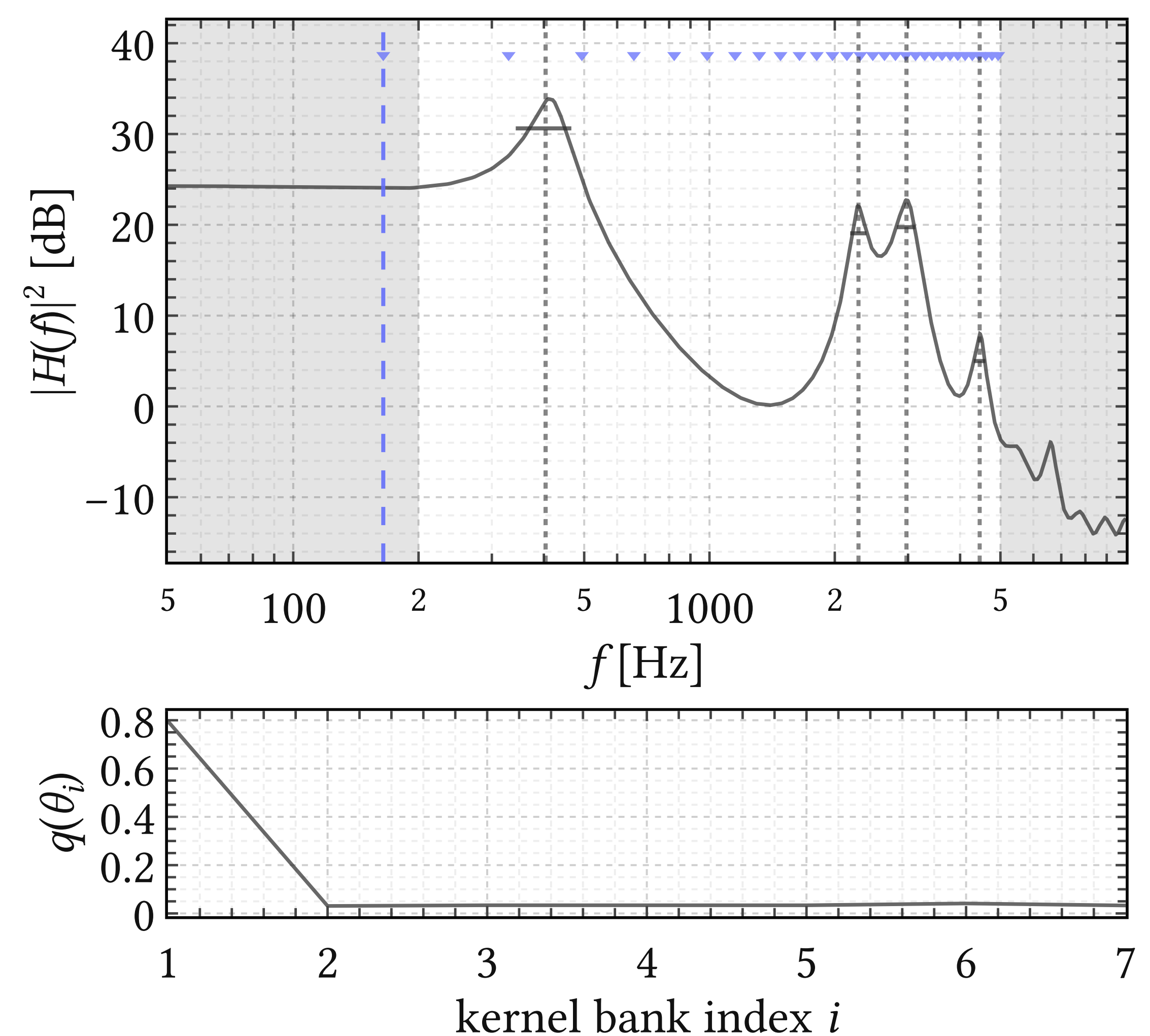
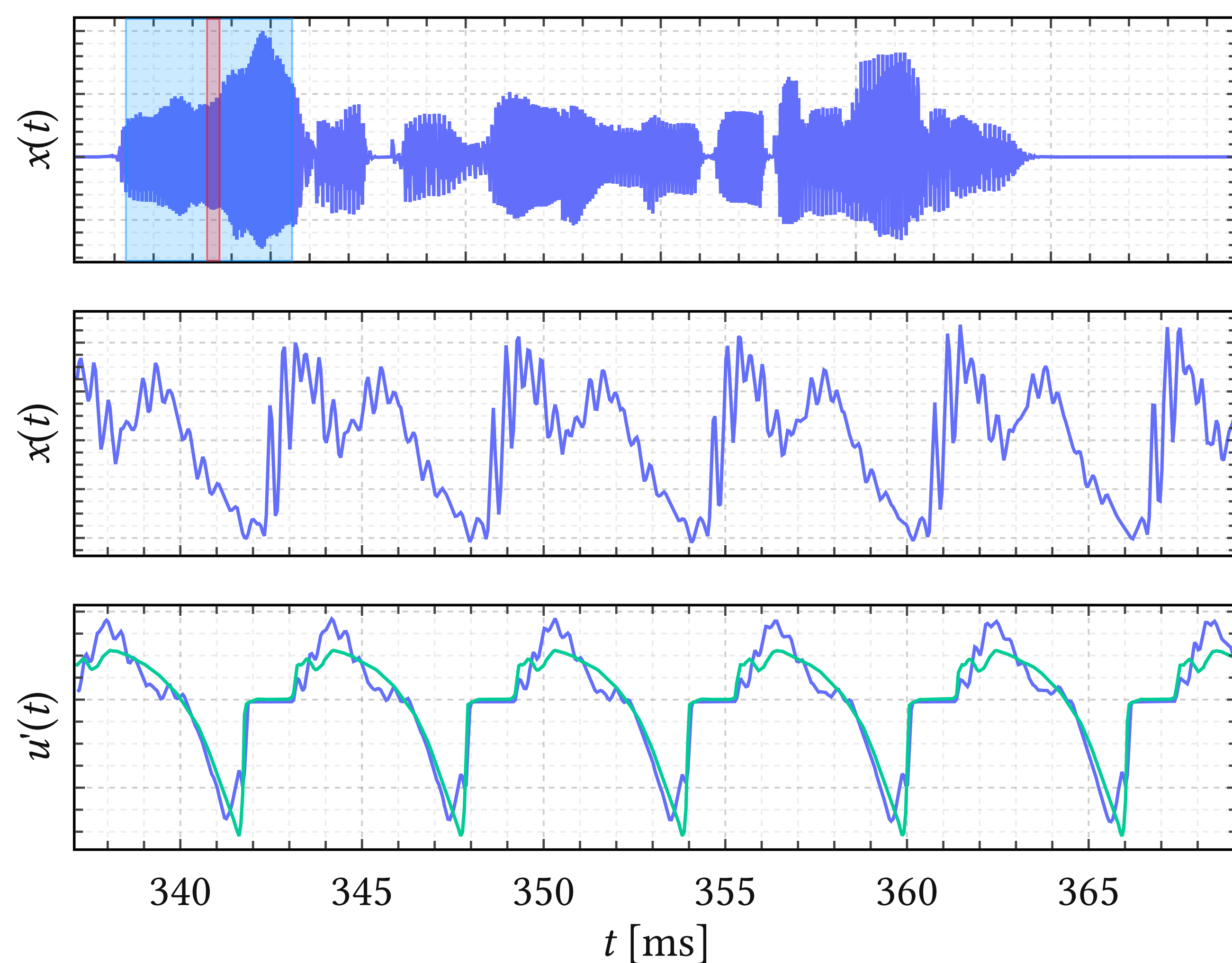


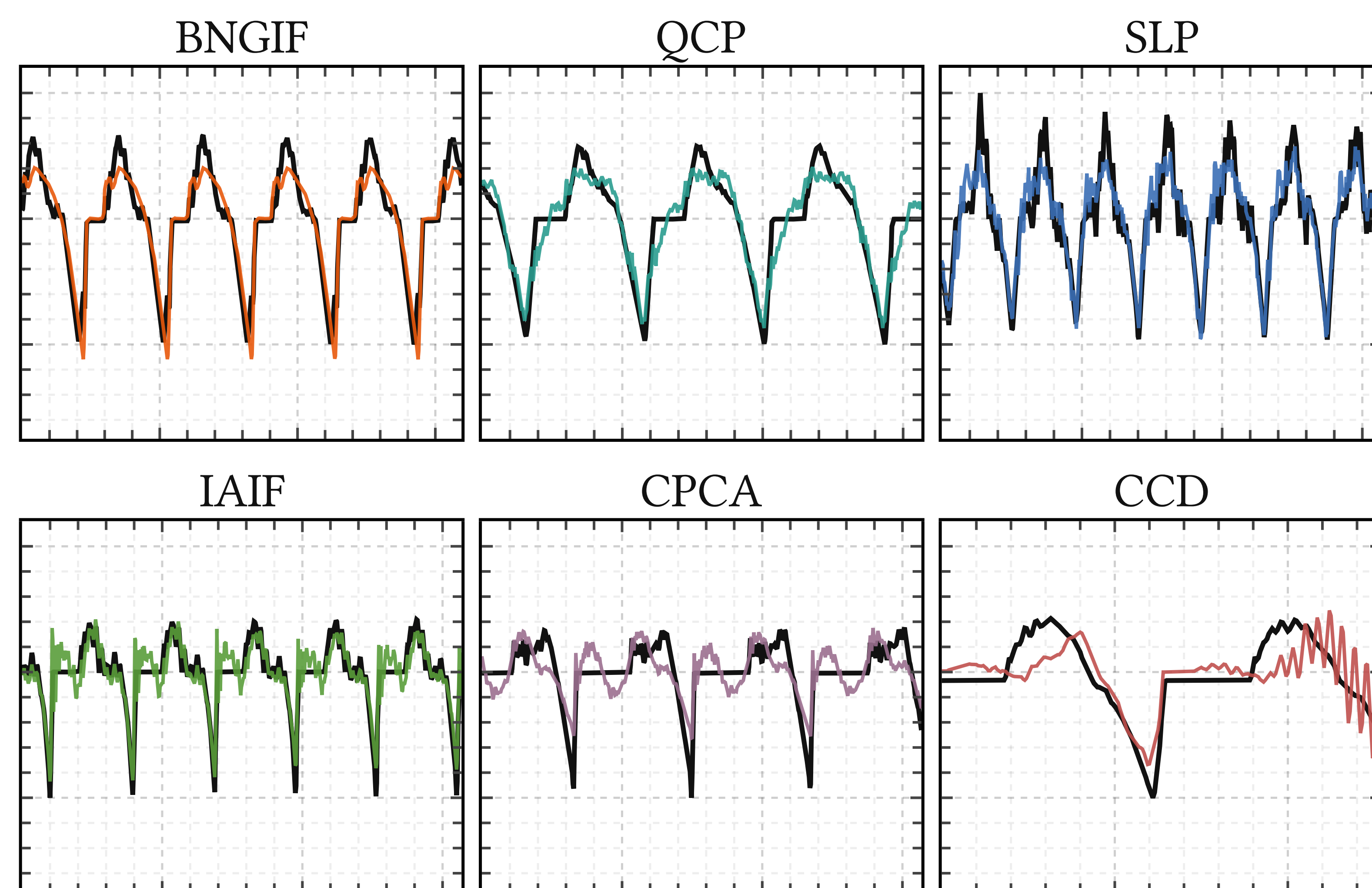
PROBABILISTIC SPEECH DECONVOLUTION IN REAL TIME

METHOD

One Bayesian model per 32 ms frame: a posterior over the glottal airflow (band), the vocal tract filter, and which prior shape explains the frame. A second of speech in 0.2 s on one GPU.



BENCHMARK



True airflow (black) and six methods' estimates on the EGIFA benchmark.

	BNGIF	QCP	SLP	IAIF	CPCA	CCD
BNGIF	0.935	<.001	<.001	<.001	<.001	<.001
QCP	1.000	0.910	<.001	<.001	<.001	<.001
SLP	1.000	1.000	0.889	<.001	<.001	<.001
IAIF	1.000	1.000	1.000	0.886	<.001	<.001
CPCA	1.000	1.000	1.000	1.000	0.759	0.004
CCD	1.000	1.000	1.000	1.000	0.996	0.741

Paired Wilcoxon over the whole benchmark: better than each, $p < .001$.

APPLICATIONS

Hearing implants

Could pitch and formants *with error bars* improve sound coding?

Clinics

Physiological voice biomarkers, not black-box features.

Phonetics, forensics

Formant tracks 30% more accurate than Praat.

